

MDR SENTINEL

HOSPITAL OUTBREAK INTELLIGENCE

Raviprolu Lakshmi Sri Divya

*Department of CSE
R.M.D Engineering College
Anna University
Thiruvallur, India
raviprolu.lakshmisridivya@gmail.com*

Revathi V

*Department of CSE
R.M.D Engineering College
Anna University
Thiruvallur, India
rekhavelu2020@gmail.com*

Putha Dhatrika

*Department of CSE
R.M.D Engineering College
Anna University
Thiruvallur, India
dhatrikap07@gmail.com*

ABSTRACT

Multidrug-resistant (MDR) infections represent a growing challenge for healthcare systems worldwide. These infections often spread rapidly within hospital environments due to frequent interactions between patients, healthcare workers, and medical facilities. Traditional infection monitoring approaches rely on manual observation and delayed reporting mechanisms, which can limit the ability of healthcare administrators to detect potential outbreaks at an early stage. As hospitals increasingly adopt digital technologies, there is a growing need for intelligent systems capable of monitoring infection risks and providing real-time insights.

This paper presents MDR Sentinel, an intelligent infection monitoring platform that integrates Bluetooth Low Energy (BLE)-based proximity tracking with a web-based analytics system. The proposed platform collects interaction data between individuals and hospital environments and applies machine learning techniques to analyse potential infection exposure patterns.

The system provides real-time dashboards that allow hospital administrators and infection control teams to identify high-risk contacts, detect infection hotspots, and monitor interaction trends across hospital wards. By automating infection monitoring and enabling data-driven decision-making, MDR Sentinel aims to assist healthcare professionals in improving infection surveillance and reducing the risk of hospital-acquired infections.

Keywords— Infection Monitoring, Bluetooth Low Energy (BLE), Machine Learning, Contact Tracing, Smart Healthcare Systems.

1. INTRODUCTION

Healthcare-associated infections remain one of the major

concerns in modern medical environments, particularly due to the increasing presence of multidrug-resistant (MDR) pathogens. These infections can spread rapidly within hospitals because of close interactions between patients, healthcare workers, and hospital infrastructure. Early detection and monitoring of infection transmission are therefore essential for preventing outbreaks and ensuring patient safety.

Traditional infection surveillance systems rely heavily on manual observation, clinical reporting, and retrospective analysis of patient records. While these methods provide useful information, they are often slow and inefficient in identifying infection sources and tracking complex contact patterns. As hospital environments become more interconnected and data-driven, there is a growing need for automated systems capable of continuously monitoring interactions and detecting potential infection risks.

Recent advancements in Internet of Things (IoT) technologies and data analytics have created new opportunities for intelligent healthcare monitoring systems. Bluetooth Low Energy (BLE)-based proximity sensing allows devices to detect nearby interactions between individuals and objects. When combined with machine learning and data analysis techniques, such systems can provide valuable insights into infection transmission patterns.

Motivated by these developments, this project proposes MDR Sentinel, a smart infection monitoring platform that integrates BLE-based contact tracking with a web-based analytics dashboard. The system collects and analyses interaction data to identify potential infection risks and provide monitoring capabilities for hospital administrators and infection control teams. By improving the visibility of contact networks and exposure events, MDR Sentinel aims to enhance hospital infection surveillance and support proactive healthcare management.

2. LITERATURE REVIEW

2.1 Smart Healthcare Monitoring Systems

The increasing integration of digital technologies into healthcare has led to the development of smart monitoring systems capable of improving patient care and hospital management. These systems often rely on sensors, wireless communication technologies, and data analytics to collect and analyse healthcare-related information in real time. Researchers have explored the use of Internet of Things (IoT) devices to monitor patient activity, environmental conditions, and healthcare workflows.

Several studies have highlighted the potential of sensor-based monitoring systems in healthcare environments. Wireless sensor networks can track patient movement, monitor environmental conditions, and detect abnormal events. Such technologies enable healthcare providers to collect continuous data without requiring manual intervention. However, many existing systems focus primarily on patient monitoring rather than infection surveillance and outbreak analysis.

2.2 Contact Tracing and Infection Monitoring Technologies

Contact tracing has become an important tool for controlling infectious disease outbreaks in healthcare settings. Digital contact tracing systems use wireless communication technologies such as Bluetooth and RFID to detect interactions between individuals. These technologies allow healthcare organizations to track contact events and identify individuals who may have been exposed to infections.

Bluetooth Low Energy (BLE) has gained significant attention as a reliable technology for proximity detection due to its low power consumption and compatibility with modern devices. BLE-based systems can detect nearby devices and record interaction events, making them suitable for monitoring contact networks within hospital environments. Researchers have demonstrated that proximity tracking can assist in identifying infection clusters and preventing further transmission.

2.3 Data Analytics in Healthcare Surveillance

Data analytics and machine learning techniques are increasingly applied to healthcare surveillance systems. By analysing large volumes of interaction data, these techniques can identify patterns, predict potential risks, and support decision-making processes. Machine learning algorithms can detect unusual interaction patterns and highlight potential infection hotspots within hospital environments.

Several studies suggest that combining proximity tracking technologies with data analytics can significantly enhance infection monitoring systems. However, many existing solutions lack integrated platforms that provide real-time visualization and user-friendly interfaces for healthcare administrators.

2.4 Research Gap and Motivation

Although numerous studies have explored IoT-based healthcare monitoring and digital contact tracing systems, there is limited research on integrated platforms that combine proximity tracking, data analytics, and real-time visualization specifically for hospital infection surveillance. Existing systems often focus on isolated components rather than providing a comprehensive monitoring solution.

This research aims to address this gap by developing MDR Sentinel, a unified platform that integrates BLE-based proximity tracking with machine learning analysis and interactive dashboards. The system is designed to support healthcare professionals in monitoring infection risks and enabling timely decision-making for outbreak prevention.

3. METHODOLOGY

3.1 Research Question

Healthcare institutions generate large volumes of interaction data as patients, healthcare workers, and hospital resources continuously interact within clinical environments. Identifying potential infection exposure within such complex interaction networks is a challenging task when performed manually. Traditional surveillance approaches often struggle to process large amounts of interaction data efficiently.

This project investigates whether an intelligent monitoring system can assist healthcare administrators in identifying infection risks more effectively. The study therefore addresses the following research question:

Can an intelligent monitoring platform that integrates BLE-based proximity tracking and machine learning analysis effectively detect infection exposure patterns within hospital environments?

The research explores whether automated interaction monitoring and data analytics can provide valuable insights that support the early detection of infection risks.

3.2 System Environment and Users

The proposed MDR Sentinel system is designed for use in hospital environments where infection monitoring and surveillance are essential. The primary users of the system include hospital administrators, infection control teams, and healthcare professionals responsible for maintaining patient safety.

The platform provides a web-based interface that allows users to access infection monitoring dashboards, analyse interaction data, and identify potential infection risks. Administrators are responsible for managing system settings and monitoring hospital-wide data, while healthcare professionals use the system to track interactions and evaluate potential exposure risks.

3.3 Data Collection and Processing

The MDR Sentinel platform collects interaction data using BLE-based proximity detection devices. These devices detect nearby interactions between individuals within hospital environments and record relevant information such as timestamps, device identifiers, and interaction duration.

Once the interaction data is collected, it is transmitted to the central system where it is processed and stored within a database. Data preprocessing techniques are applied to organize and clean the collected data to ensure accurate and reliable analysis.

Machine learning algorithms are then applied to analyse interaction patterns and estimate potential infection risk levels. The system identifies possible exposure events and highlights individuals who may have had high-risk interactions. The analysed data is presented to users through an interactive dashboard that enables administrators to monitor infection trends and identify potential outbreak clusters.

4. SYSTEM ARCHITECTURE

4.1 Technical Architecture

The MDR Sentinel system follows a modular architecture designed to support efficient data collection, processing, and analysis. The architecture consists of several components including the data collection module, processing engine, and visualization interface (*Refer Fig.1*).

The data collection module gathers interaction data from BLE-based proximity devices within hospital environments. The collected data is transmitted to a central processing system where it is organized and stored within a structured database. The analysis module applies machine learning algorithms to detect interaction patterns and identify potential infection risks.

Finally, the visualization module presents the analysed data through a web-based dashboard that allows healthcare administrators to monitor infection trends and exposure events efficiently.

4.2 System Workflow

The system workflow describes the sequence of operations involved in monitoring infection interactions. The process begins when BLE devices detect proximity interactions between individuals within hospital environments. These interactions are recorded and transmitted to the central system.

The collected data is then processed and stored in a database where it can be analysed using machine learning techniques. The analysis module identifies interaction patterns and estimates infection exposure levels based on proximity and duration of contact.

Users can access the system through the web dashboard to view interaction data, monitor potential infection risks, and analyse historical trends. The system provides visual insights that assist administrators in making informed decisions regarding infection control measures.

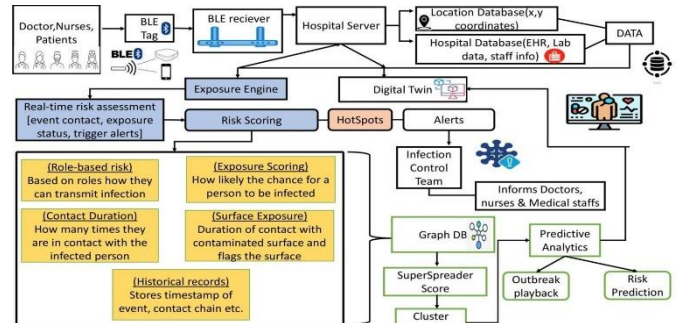


Fig 1: Flow Diagram of MDR Sentinel

5. EXPECTED RESULTS

The MDR Sentinel system was implemented as a prototype web-based monitoring platform designed to visualize interaction patterns within simulated hospital environments.. The system integrates BLE-based proximity detection with a centralized data analytics platform to support infection risk monitoring.

The developed dashboard enables healthcare administrators to view interaction statistics, identify potential high-risk contacts, and monitor possible infection clusters within hospital wards. The platform presents visual representations of interaction networks and exposure events, allowing users to analyse infection patterns more effectively (*Refer Fig.2*).

The system demonstrates the feasibility of using proximity tracking and data analytics to support improved infection monitoring and surveillance in healthcare environments.

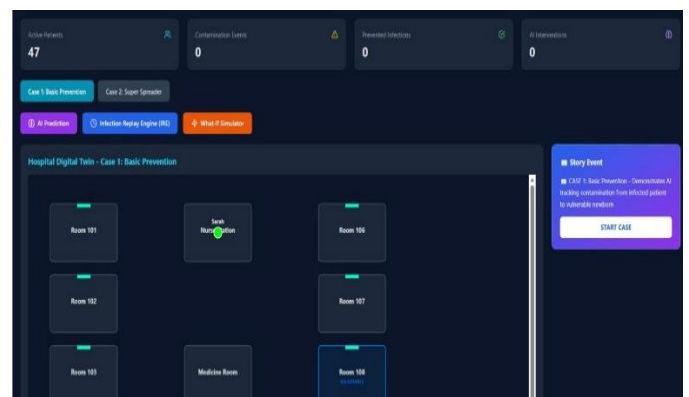


Fig. 2: ICT Real-Time Monitoring Dashboard

6. MARKET ANALYSIS AND DEPLOYMENT POTENTIAL

The MDR Sentinel platform has strong potential for deployment within modern healthcare systems. Hospitals increasingly require intelligent monitoring solutions that can improve infection surveillance and enhance patient safety.

A market analysis was conducted using the TAM–SAM–SOM framework to estimate the potential adoption of infection monitoring technologies (*Refer Fig.3*).

- TAM: Represents the total healthcare market that could benefit from infection monitoring solutions.
- SAM: Represents hospitals capable of implementing digital monitoring infrastructure.
- SOM: Represents the realistic portion of the market targeted during early deployment.

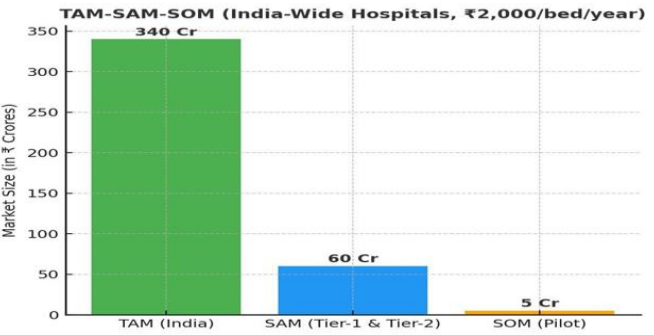


Fig. 3: TAM–SAM–SOM Market Opportunity Analysis for MDR SENTINEL

7. APPLICATIONS

The MDR Sentinel system can be applied in several healthcare scenarios to improve infection monitoring and hospital safety. One of the primary applications is real-time tracking of patients, doctors, and hospital staff using Bluetooth Low Energy (BLE) technology, which helps monitor interactions and identify potential exposure events within hospital environments.

Another important application is infection risk prediction and outbreak detection (*Refer Fig.4*). By analysing contact patterns and movement data using machine learning techniques, the system can identify high-risk contacts and assist in identifying potential infection risks and outbreak patterns within hospital environments.

The system also supports digital twin–based hospital visualization (*Refer Fig.5*), where a virtual model of the hospital environment is created to simulate patient movement

and infection spread. This enables hospital administrators to analyse infection transmission patterns and identify high-risk zones for proactive intervention.

Additionally, MDR Sentinel provides automated alert systems and infection replay analysis. Real-time alerts notify infection control teams about potential exposure events, while the infection replay feature helps analyse past infection spread patterns to determine root causes and improve preventive strategies.

Overall, MDR Sentinel can be used to enhance hospital infection surveillance, decision support, and proactive outbreak prevention, thereby improving patient safety and healthcare efficiency.

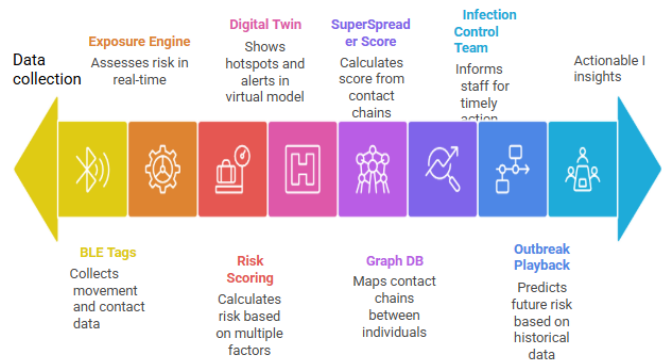


Fig. 4: From data collection to actionable insights for infection control

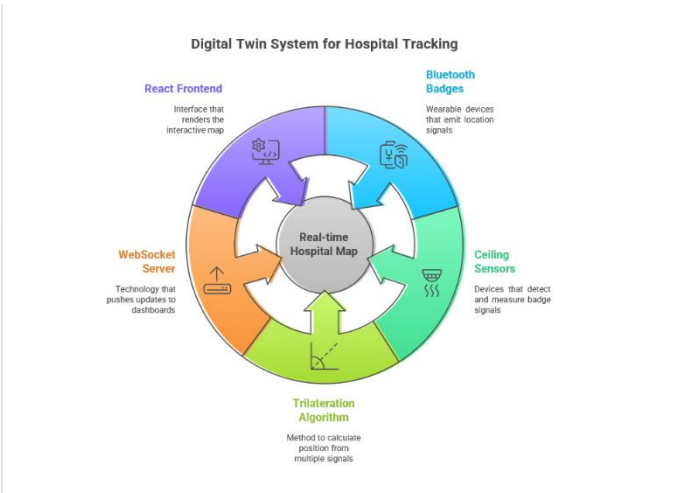


Fig 5: Digital Twin System for Hospital Tracking

8. CONCLUSION

In order to proactively identify, track, and prevent Multi-Drug Resistant (MDR) infections in hospital settings, the proposed work introduces MDR Sentinel: Hospital Outbreak Intelligence, an AI-driven smart surveillance framework. The system overcomes data fragmentation and delayed response in conventional infection control. The proposed framework demonstrates how clinical records, interaction data, and environmental information could be integrated into a unified infection monitoring platform.

The system uses Bluetooth Low Energy (BLE) technology via mobile devices and BLE-enabled tags affixed to patients, medical personnel, and medical equipment to enable real-time proximity tracking and automated contact tracing. This makes it possible to continuously observe how people interact and move around hospital areas. This multi-source data is analyzed by machine learning-based predictive analytics models to find early infection clusters, high-risk areas, and possible outbreak trends prior to widespread transmission occurs.

MDR Sentinel further provides interactive dashboards and real-time visualization tools that deliver predictive alerts, risk scores, and actionable insights to infection control teams and hospital administrators. By transforming infection surveillance from a reactive approach to a proactive, data-driven, and technology-enabled system, the proposed work supports faster containment, improved operational efficiency, regulatory compliance, and enhanced patient safety across hospital settings.

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